

Management Information Systems

Class Notes: Classes 14-23

Part II • Sections MIS-1, MIS-2, MIS-3 • Fundamentals, IS Concepts, IS & Organization

CO1 — MIS Role in Organizational Performance

CO2 — MIS in Business Strategy & Decision Making

Bloom's: Remember → Analyze

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1. Definition of MIS

CORE DEFINITION

Management Information System (MIS) is an integrated, user-machine system for providing information to support the operations, management, analysis, and decision-making functions in an organization. It makes use of computer hardware, software, manual procedures, models for analysis, planning, control and decision-making, and a database.

A simpler working definition: MIS is an integrated system of people, processes, data, and technology that collects, processes, stores, and disseminates information to support management decision-making and organizational performance.

BREAKING DOWN THE DEFINITION

- ▶ **Integrated:** MIS brings together data from multiple departments (finance, HR, sales, operations) into a single coherent system — avoiding information silos.
- ▶ **User-machine system:** Humans and computers work together. Computers handle data processing; humans provide judgment and decision-making.
- ▶ **Supports operations and management:** MIS serves all levels — operational staff (daily transactions), middle managers (monitoring and control), and top executives (strategic planning).
- ▶ **Database:** A structured repository of organizational data that the MIS draws from to generate information.
- ▶ **Models:** Mathematical or analytical models embedded in MIS for forecasting, optimization, or simulation.

KEY INSIGHT

MIS is not just software. It is a socio-technical system — it encompasses people (managers, analysts, IT staff), processes (reporting workflows, approval chains), technology (hardware, software, networks), and data (databases, documents). Removing any one element breaks the system.

2. Characteristics of MIS

A well-designed MIS has the following distinguishing characteristics:

Goal-Oriented

Integrated

MIS is designed around organizational goals. Every report, dashboard, or alert is tied to a specific management objective — e.g., reducing inventory costs, improving customer response time.

MIS crosses functional boundaries. A sales report may pull data from the inventory system, the customer database, and the financial ledger simultaneously — providing a holistic view.

Flexible

MIS can adapt to changing information needs. Managers can request ad-hoc reports, change report formats, or add new data dimensions without redesigning the entire system.

Reliable

MIS must produce consistent, accurate results. The same query run at the same time should always produce the same output. Data integrity controls ensure this.

Timely

Information must be available when needed. A daily sales report delivered a week late is useless. MIS is designed for appropriate timeliness — real-time for operational systems, daily/weekly for management reports.

Cost-Effective

The value of information produced by MIS must exceed the cost of producing it. This is measured through ROI analysis: if a ₹5 lakh system saves ₹20 lakh in decision errors annually, it is cost-effective.

3. MIS Services

MIS provides four core services to management:

- ▶ **Reporting:** Scheduled, exception, and demand reports summarizing operations. Example: a weekly inventory status report showing stock levels vs. reorder points.
- ▶ **Query Processing:** Managers can ask ad-hoc questions and get answers from the database. Example: "How many units of Product X were sold in Dhaka last month?"
- ▶ **Exception Alerting:** MIS automatically flags conditions that deviate from norms. Example: an alert when machine downtime exceeds 5%, or when a customer account is overdue by 60 days.
- ▶ **Forecasting Support:** MIS uses historical data and models to project future conditions. Example: sales forecasting based on seasonal trend analysis.

4. Limitations of MIS

- ▶ **Cannot handle unstructured problems:** MIS is designed for structured and semi-structured decisions. For highly ambiguous, creative, or political decisions (e.g., "Should we enter a new market?"), MIS provides only partial support.

- ▶ **GIGO — Garbage In, Garbage Out:** If the input data is inaccurate, incomplete, or outdated, the MIS output will be equally flawed. Data quality is the foundation of any MIS.
- ▶ **High cost of development:** Building an enterprise MIS (e.g., SAP, Oracle ERP) can cost millions of dollars and years to implement. SMEs may find this prohibitive.
- ▶ **Resistance to change:** Employees may resist adopting MIS due to fear of job loss, lack of training, or comfort with existing manual systems.
- ▶ **Security and privacy risks:** Centralizing organizational data creates a single point of failure. A breach can expose sensitive business and customer information.
- ▶ **Inflexibility of rigid systems:** Older, packaged MIS may not adapt well to unique organizational processes, requiring costly customization.

EXAM TIP

"GIGO" is a favourite exam concept. Always explain it with an example: if a retail store enters wrong sales figures (e.g., 100 units instead of 10), the MIS will generate a misleading demand forecast, leading to overstock. The system is only as good as the data fed into it.

5. MIS vs. Data Processing System (DPS)

Dimension	Data Processing System (DPS)	Management Information System (MIS)
Primary Focus	Processing day-to-day transactions efficiently	Providing information for managerial decisions
Output	Transaction records, receipts, ledgers	Summarized reports, dashboards, exception alerts
Users	Operational-level clerical staff	Middle and top management
Time horizon	Current / real-time transactions	Historical trends + current + some future forecasts
Example	Payroll processing, invoicing, order entry	Monthly sales performance report, budget variance analysis
Scope	Narrow — one function at a time	Broad — integrates data from multiple functions

6. ISO/IEC 27001 — Information Management Framework

STANDARD REFERENCE

ISO/IEC 27001 is the international standard for Information Security Management Systems (ISMS). It provides a framework for establishing, implementing, maintaining, and continually improving an organization's information security — directly relevant to any MIS deployment.

Key requirements of ISO/IEC 27001 relevant to MIS:

- ▶ **Risk Assessment:** Identify information assets and assess threats and vulnerabilities.
- ▶ **Access Control:** Only authorized users access specific information — role-based access in MIS.
- ▶ **Information Classification:** Label data as public, internal, confidential, or restricted.
- ▶ **Incident Management:** Procedures for responding to data breaches or system failures.
- ▶ **Business Continuity:** Ensure MIS remains operational (or recoverable) after disruptions.

1. Types of MIS – Matched to Management Levels

MIS is not a single monolithic system. Different types of MIS serve different levels of the management hierarchy. The classic model maps three MIS types to three management levels:

Strategic MIS

Top Management

Tactical MIS

Middle Management

Operational MIS

Lower / Operational Level

MIS Type	Management Level	Time Horizon	Decision Type	Example Output
Operational MIS	Supervisors, team leads	Daily / weekly	Structured, routine	Daily sales transaction report, attendance log, inventory status
Tactical MIS	Department managers, middle management	Weekly / monthly / quarterly	Semi-structured	Monthly budget variance report, departmental performance dashboard
Strategic MIS	CEO, board, top executives	Annual / multi-year	Unstructured	5-year market trend analysis, competitor benchmarking report

2. Transaction Processing System (TPS)

DEFINITION

A **Transaction Processing System (TPS)** is a computerized system that processes routine, high-volume business transactions in a reliable, accurate, and timely manner. TPS is the operational backbone of an organization — it generates the raw data that feeds higher-level MIS.

TPS PROCESSING MODES

Batch Processing

Transactions are collected and processed together at scheduled intervals (e.g., end of day, end of week). Suitable for non-urgent, high-volume tasks.

Examples: weekly payroll processing, monthly bank statement generation, end-of-day inventory update.

Real-Time Processing

Each transaction is processed immediately as it occurs. Provides instant feedback. Necessary when up-to-date information is critical.

Examples: ATM cash withdrawal, airline seat reservation, stock trade execution, hospital patient admission.

KEY PROPERTIES OF TPS

- ▶ **Atomicity:** A transaction is all-or-nothing. If a bank transfer fails midway, the entire operation is rolled back — money neither disappears nor is duplicated.
- ▶ **Consistency:** The database moves from one valid state to another valid state after each transaction.
- ▶ **Isolation:** Concurrent transactions do not interfere with each other.
- ▶ **Durability:** Once committed, a transaction's changes persist even if the system crashes. (These four together = ACID properties — important for CSE students.)

CSE CONNECTION — ACID PROPERTIES

ACID (Atomicity, Consistency, Isolation, Durability) is a concept you study in Database Management Systems. In an MIS context, ACID guarantees that TPS transactions are reliable. A payroll system must be ACID-compliant — an employee must never be paid twice due to a system failure, and a failed run must not corrupt the payroll database.

3. Management Reporting System (MRS)

The Management Reporting System (MRS) is the core of tactical MIS. It processes TPS data and generates structured reports for middle managers.

TYPES OF MRS REPORTS

- ▶ **Scheduled Reports:** Generated at fixed intervals regardless of content. Example: daily sales report emailed to the sales manager at 8 AM every day.
- ▶ **Exception Reports:** Generated only when something deviates from a threshold. Example: a report triggered only when inventory drops below the reorder level, or when an employee's leave balance exceeds 30 days. This focuses management attention on problems — not routine data.
- ▶ **Demand Reports (Ad-hoc Reports):** Generated on request by a manager. Example: a manager asks, "Show me all customers who have not placed an order in 90 days." The system queries the database and

produces the report immediately.

WORKED EXAMPLE — EXCEPTION REPORT

A garment factory sets a quality threshold: reject rate must not exceed 2%. The MRS monitors the quality TPS data. On Tuesday, Line 3 shows a 4.7% reject rate. The MRS immediately generates an exception report and sends it to the production manager. Without the exception report, the manager would have to manually scan hundreds of lines of production data — the exception report saves time and focuses attention where needed.

4. Importance of MIS

- ▶ **Improved Decision Speed:** Managers receive summarized, relevant information instantly. A decision that once took days (waiting for manual reports) can now be made in minutes with a dashboard.
- ▶ **Competitive Advantage:** Organizations with better MIS make faster, more accurate decisions than competitors. Amazon's real-time inventory and pricing MIS is a core competitive weapon.
- ▶ **Process Automation:** MIS automates repetitive tasks — payroll calculation, report generation, invoice processing — freeing human effort for higher-value work.
- ▶ **Data Centralization:** Instead of each department maintaining separate, inconsistent records, MIS creates a single source of truth for the entire organization.
- ▶ **Performance Monitoring:** MIS enables real-time tracking of KPIs (Key Performance Indicators) — managers know immediately if targets are being met.
- ▶ **Cost Reduction:** Automation reduces labour costs; better inventory management reduces holding costs; better demand forecasting reduces stockouts and overstock.

5. Potential Risks of Information Systems

- ▶ **System Failures:** Hardware crashes, software bugs, or network outages can halt operations. A bank whose core banking system goes down cannot process transactions.
- ▶ **Data Breaches:** Centralized MIS databases are attractive targets for hackers. A breach may expose customer data, trade secrets, or financial records.
- ▶ **Over-reliance:** When MIS goes down, staff may not know how to function manually. Organizations become brittle without fallback procedures.
- ▶ **Change Resistance:** Employees may resist MIS adoption, leading to underutilization. A ₹10 crore ERP system that staff bypass using spreadsheets delivers no ROI.

- ▶ **Data Quality Degradation:** GIGO — poor data entry practices corrupt the MIS database over time, leading to flawed decisions.
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- ▶ **Privacy Concerns:** MIS tracks employee and customer behaviour in detail, raising ethical concerns about surveillance and consent.

6. MIS Success Factors

- ▶ **Top Management Support:** Without commitment from leadership (budgetary, cultural, strategic), MIS projects fail. The CIO must report to the CEO level.
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- ▶ **User Involvement:** MIS designed without input from actual users produces systems that don't match real workflows. Involve department heads and front-line staff in requirements gathering.
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- ▶ **Clear Objectives:** MIS must have defined goals — which decisions it will support, which KPIs it will track, which reports it will produce.

1. Data — Definition

DEFINITION

Data are raw, unprocessed facts — numbers, characters, symbols, images, or events — recorded without context or interpretation. Data on its own does not carry meaning or enable decision-making.

Example: The string 47, 63, 52, 88, 71 is data. Without context, it is meaningless — it could be temperatures, scores, ages, or prices.

2. Information — Definition

DEFINITION

Information is data that has been processed, organized, structured, or presented in a context that makes it meaningful and useful for decision-making. Information answers a question or reduces uncertainty.

Example (same data): "The exam scores of 5 students in CS101 are: 47, 63, 52, 88, 71. The class average is 64.2, and the highest score is 88." Now the data has become information — a teacher can identify struggling students and plan interventions.

3. Comparison: Data vs. Information

Dimension	Data	Information
Nature	Raw, unprocessed facts	Processed, meaningful output
Context	No context — standalone facts	Has context — answers a question
Format	Numbers, text, images, audio (raw)	Reports, summaries, charts, analyses
Purpose	Input for processing	Basis for decisions
Dependence	Independent — exists before processing	Dependent — derived from data
Meaning	Meaningless without context	Meaningful and actionable

Dimension	Data	Information
Example	Temperature readings: 36.8, 37.2, 38.5, 39.1 °C	"Patient's temperature has risen 2.3°C in 24 hours — possible fever developing."
Decision-making	Cannot support decisions directly	Directly supports decisions

4. The DIKW Hierarchy

The DIKW model extends the data–information distinction into a four-level hierarchy of understanding:

Wisdom	Knowing what to do. Applying knowledge with judgment, experience, and ethics. Example: A seasoned CEO uses past knowledge of market cycles to avoid over-expanding in a boom — knowing it will be followed by a bust. <i>"Why?" and "What should we do?"</i>
Knowledge	Understanding patterns and principles. Information organized into frameworks, rules, and expertise. Example: A doctor knows that a rising temperature + elevated white blood cell count = likely infection — this pattern is knowledge. <i>"How does this work?"</i>
Information	Processed data with context and meaning. Example: "Sales in Region A fell 15% in Q3 compared to Q2." This is information — it contextualizes the raw sales figures. <i>"Who? What? When? Where?"</i>
Data	Raw facts without context. Example: sales figures 4500, 3825, 4100 for three months. <i>No inherent meaning alone.</i>

EXAM TIP

DIKW hierarchy questions are common. You must be able to: (1) define each level, (2) give an original example for each level, (3) explain how each level transforms into the next. A common trap: many students confuse "knowledge" with "information." Knowledge = internalized patterns + experience; information = structured data output.

5. Quality Attributes of Good Information

For information to be useful in management decision-making, it must meet the following quality criteria:

Accurate	Timely
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Free from errors. If a sales report overstates revenue by 10%, the manager may approve bonuses that aren't deserved, straining the budget.

Available when needed. A stock-out alert generated 3 days after the stockout is worthless. Information has a "freshness" window beyond which its value decays.

Complete

Contains all necessary facts. A financial report missing the expenses column gives a misleadingly positive picture of profitability.

Relevant

Pertains to the decision at hand. A marketing manager receiving a detailed factory maintenance report is getting irrelevant information — it wastes time and clutters decision-making.

Accessible

Available to authorized users when needed, in a usable format. A report that takes 4 hours to generate and requires IT department intervention is poorly accessible.

Consistent

Does not contradict other information in the system. If the HR system shows 500 employees but the payroll system shows 490, there is an inconsistency — which figure do managers trust?

6. Cost of Poor-Quality Information

RESEARCH FINDING — MCKINSEY INSTITUTE

Research by the McKinsey Global Institute estimates that organizations lose **15–25% of productivity** due to poor-quality information — time spent searching for data, correcting errors, reconciling conflicting reports, and making decisions based on flawed information. For a 1,000-person organization, this can represent hundreds of person-years of wasted effort annually.

CONSEQUENCES OF POOR INFORMATION QUALITY

- ▶ **Wrong decisions:** A manager approves a supplier based on a credit report with outdated data — the supplier has since gone bankrupt.
- ▶ **Compliance failures:** Incorrect financial data leads to wrong tax filings or regulatory reports — resulting in fines.
- ▶ **Customer dissatisfaction:** A customer is billed the wrong amount due to erroneous pricing data in the system.
- ▶ **Resource waste:** A factory orders excess raw materials because the inventory system double-counted existing stock.

7. Information Overload

DEFINITION

Information overload occurs when the volume of information presented to a decision-maker exceeds their cognitive capacity to process it effectively, leading to delayed decisions, poor choices, or decision paralysis.

CAUSES

- ▶ Too many reports with too much detail — managers receive 100-page weekly reports when they need a 1-page summary
- ▶ Poorly filtered MIS — the system sends every exception alert, not just critical ones
- ▶ Multiple overlapping systems generating duplicate or conflicting data
- ▶ Email and notification culture — constant interruptions fragment attention

MANAGEMENT STRATEGIES

- ▶ **Exception reporting:** Only surface data that exceeds defined thresholds
- ▶ **Dashboards with drill-down:** Show high-level KPIs; let managers choose when to dig deeper
- ▶ **Role-based filtering:** Each manager sees only information relevant to their function and level
- ▶ **Prioritization frameworks:** Use urgency × importance matrices to rank incoming information

1. The Data Processing Cycle

Converting raw data into usable information follows a systematic cycle:

Stage	What Happens	Example (Sales System)
1. Collection	Raw data is gathered from sources	Cashiers scan items at POS terminals; online orders submitted
2. Input	Data is entered into the system (validated)	Transaction records written to the sales database
3. Processing	Data is transformed via operations	Total revenue calculated; returns subtracted; discounts applied
4. Storage	Processed data stored for future use	Daily sales stored in data warehouse by region, product, date
5. Output	Information presented to users	Monthly regional sales report generated for management
6. Feedback	Output used to refine input or processing	Manager identifies error in discount calculation; data re-processed

2. Data Processing Operations

Within the "Processing" stage, several fundamental operations transform data:

- ▶ **Classifying:** Assigning data to categories. Example: classifying sales transactions by product category (Electronics, Clothing, Food) or by region (North, South, East, West).
- ▶ **Sorting:** Arranging data in a defined sequence. Example: sorting customer orders by date, or employee records alphabetically. Sorting makes data searchable and comparable.
- ▶ **Calculating:** Performing arithmetic operations on data. Example: computing total revenue (units × price), calculating profit margins, computing average order value.
- ▶ **Summarizing:** Aggregating detailed data into summary form. Example: 10,000 daily sales transactions → "Total monthly revenue: \$42,50,000." Managers need summaries, not raw transaction logs.
- ▶ **Storing:** Persisting processed data for later retrieval. Includes indexing and organizing data for efficient access.

- ▶ **Retrieving:** Extracting specific data from storage on demand. Example: querying the database for "all invoices over 1,00,000 from the last 90 days."

3. Input Validation

Before data is processed, it must be validated — checked for accuracy and completeness. Poor validation = GIGO. Common validation checks:

Range Check

Ensures a value falls within acceptable bounds.

Example: Age must be between 0 and 120. An entry of 999 or -5 fails the range check. Exam score must be 0–100.

Type Check

Ensures data matches the expected data type.

Example: A phone number field must contain only digits. A name field must contain only letters. Entering "abc" into an age field fails type check.

Presence Check

Ensures a required field is not left empty.

Example: A customer order form cannot be submitted without a delivery address. Employee records must have a national ID number.

Consistency Check

Ensures two related fields are logically consistent.

Example: "Date of joining" cannot be earlier than "Date of birth." "Order delivery date" cannot be before "Order date."

4. ETL Process — Data Warehouses

DEFINITION

ETL (Extract, Transform, Load) is the process pipeline used to move data from operational source systems (TPS, ERP) into a data warehouse or analytical MIS. It is the backbone of enterprise-scale MIS and Business Intelligence (BI) systems.

○ Extract

Data is pulled from multiple source systems — the sales TPS, the HR system, the inventory database, external market data APIs. The extract process handles different formats, database types (Oracle, MySQL, PostgreSQL), and schedules (real-time vs. nightly batch).

○ Transform

Raw data is cleaned, standardized, and enriched. This is the most complex step: removing duplicates, handling missing values, converting currencies, standardizing date formats (DD/MM/YYYY vs. MM/DD/YYYY), mapping product codes across systems, and applying business rules (e.g., calculating gross margin from revenue and cost).

○ Load

Transformed data is loaded into the data warehouse or MIS database, ready for reporting and analysis. Loading may be full (replace all data) or incremental (only new/changed records).

CSE CONNECTION — ETL IN PRACTICE

ETL pipelines are built using tools like Apache Spark, Apache Airflow, Talend, Microsoft SSIS, or custom Python scripts. A CSE student implementing an MIS project would typically build an ETL pipeline that reads from a MySQL transactional database, transforms data using Python (pandas), and loads it into a PostgreSQL analytical schema. This is exactly what industry data engineers do at companies like Pathao, bKash, and Chaldal in Bangladesh.

5. Signal-to-Noise Ratio in Information

$$SNR = \text{Useful Information} \div \text{Total Data Volume}$$

Borrowed from signal processing, the **Signal-to-Noise Ratio (SNR)** in an information context measures the proportion of genuinely useful, actionable information in the total data presented to a manager.

WORKED EXAMPLE

EXAMPLE — SNR CALCULATION

A manager receives a 500-row sales report. After reviewing it, only 35 rows contain actionable insights (products underperforming, regions declining, anomalies requiring investigation).

$$SNR = 35 \div 500 = 0.07 = 7\%$$

A 7% SNR means 93% of the data in the report is noise — not directly useful. A well-designed MIS exception report would surface only those 35 rows, achieving near-100% SNR. This is why exception reports are more effective than full data dumps for management decision-making.

6. Batch Processing vs. Real-Time Processing — Trade-offs

Dimension	Batch Processing	Real-Time Processing
When processed	Collected and processed at scheduled intervals	Processed immediately as each event occurs
Data freshness	Data may be hours or days old	Data is current (seconds-level latency)

Dimension	Batch Processing	Real-Time Processing
Cost	Lower — efficient use of computing resources	Higher — requires always-on infrastructure
Complexity	Simpler to implement	Complex — must handle concurrency, failure recovery
Use cases	Payroll, monthly reports, archival backup, billing cycles	ATM transactions, fraud detection, stock trading, emergency alerts
Example	A retailer generates daily inventory reports at midnight	A bank immediately alerts the customer when a suspicious card transaction occurs

7. Practical Example: Raw Transactions → Management Report

FULL WORKED EXAMPLE

Scenario: A supermarket chain has 15 branches. Each branch's POS system generates transaction records throughout the day.

Raw data (sample): 2025-06-10, Branch-07, SKU-4421, Qty:3, Unit Price:250, Discount:10%, Payment:bKash

Step 1 — Classify: Tag each transaction by category (Grocery, Electronics, Clothing), branch region, payment method.

Step 2 — Calculate: Net revenue per transaction = $\text{Qty} \times \text{Unit Price} \times (1 - \text{Discount}\%) = 3 \times 250 \times 0.9 = \text{₹}675$

Step 3 — Summarize: Aggregate by branch and category for the week → "Branch-07: Grocery ₹1,25,000 | Electronics ₹42,000 | Clothing ₹18,500"

Step 4 — Compare: Compare against last week and same week last year → "Branch-07 grocery sales down 12% WoW — flag for manager review."

Output (Information): Regional weekly sales performance dashboard showing each branch vs. target, with exception alerts for branches more than 10% below target.

1. How MIS Supports Operational Efficiency

MIS improves operational efficiency by automating routine workflows, reducing human error, and accelerating information flow across the organization:

- ▶ **Automating routine workflows:** MIS replaces manual data entry, calculation, and report generation. A payroll system calculates salaries for 5,000 employees in minutes — a task that would take a human team weeks.
- ▶ **Reducing cycle times:** With MIS, order processing that once took 3 days (paper order → approval → warehouse → dispatch) can be reduced to hours via automated order management systems.
- ▶ **Eliminating rework:** Real-time validation prevents errors from propagating. Catching a pricing error at the point of entry is 100× cheaper than catching it after invoices are sent.
- ▶ **Enabling parallel processing:** Multiple departments access the same MIS simultaneously — the warehouse team sees an order the moment it is placed, rather than waiting for a paper copy to arrive.

2. MIS and Competitive Advantage — Porter's Value Chain

Michael Porter's Value Chain model identifies the activities a firm performs to create value. MIS can be integrated at every stage to enhance performance:

Value Chain Activity	MIS Application	Competitive Benefit
Inbound Logistics	Supplier MIS, automated purchase orders, inventory tracking	Lower procurement costs, fewer stockouts
Operations	Production scheduling MIS, quality control systems, ERP	Higher throughput, lower defect rates
Outbound Logistics	Order management systems, fleet tracking, delivery MIS	Faster delivery, lower logistics cost
Marketing & Sales	CRM systems, campaign analytics, pricing MIS	Better targeting, higher conversion rates
Service	Customer service MIS, warranty tracking, helpdesk systems	Higher customer satisfaction, lower churn

Value Chain Activity	MIS Application	Competitive Benefit
Support Activities	HR-MIS, Finance-MIS, R&D project tracking	Cost reduction, better resource allocation

3. KPIs and Dashboards

DEFINITION

A **Key Performance Indicator (KPI)** is a measurable value that demonstrates how effectively an organization is achieving its key business objectives. MIS enables real-time tracking of KPIs through dashboards — visual displays of the most critical metrics.

EXAMPLES OF KPIS BY FUNCTION

Function	KPI	MIS Role
Sales	Monthly Revenue, Conversion Rate, Average Order Value	Aggregates POS/e-commerce transaction data
Operations	Machine Uptime %, Order Fulfillment Time, Defect Rate	Monitors production sensors and quality logs
HR	Employee Turnover Rate, Absenteeism Rate, Training Hours	Tracks HR records and attendance systems
Finance	Net Profit Margin, Current Ratio, Days Sales Outstanding (DSO)	Integrates accounting system data
Customer Service	Customer Satisfaction Score (CSAT), First Call Resolution Rate	Aggregates helpdesk ticket data

4. The Balanced Scorecard (BSC)

DEFINITION

The **Balanced Scorecard (BSC)**, developed by Kaplan and Norton, is a strategic performance management framework that links organizational activities to four perspectives of performance. MIS provides the data backbone that makes BSC tracking possible.

Financial Perspective

Customer Perspective

"How do we look to shareholders?"

Tracks revenue growth, profitability, cost reduction, return on investment.

MIS provides: income statements, budget variance reports, ROI calculations

"How do customers see us?"

Tracks customer satisfaction, market share, customer retention, delivery performance.

MIS provides: CRM data, NPS scores, complaint resolution times

Internal Process Perspective

"What must we excel at?"

Tracks production efficiency, quality metrics, cycle times, innovation rate.

MIS provides: production logs, quality control data, process timing records

Learning & Growth Perspective

"Can we continue to improve?"

Tracks employee skills, IS capabilities, organizational learning.

MIS provides: training records, IS system performance metrics, knowledge base usage

5. MIS ROI – Cost-Benefit Framework

$$\text{ROI} = (\text{Net Benefit} \div \text{Cost}) \times 100\%$$

WORKED EXAMPLE – MIS ROI CALCULATION

A textile company invests ₹50,00,000 (50 lakh) in an inventory MIS system. After one year:

- Reduction in stockout losses: ₹15,00,000
- Reduction in overstock carrying costs: ₹12,00,000
- Labour savings (automated reporting): ₹8,00,000
- Improved order fulfillment (fewer errors): ₹6,00,000
- **Total Annual Benefit: ₹41,00,000**

Net Benefit = ₹41,00,000 – ₹50,00,000 = –₹9,00,000 in Year 1

Year 2 (no implementation cost): Net Benefit = ₹41,00,000; ROI = $(41,00,000 \div 50,00,000) \times 100\% = 82\%$ **ROI in Year 2**

Note: MIS investments typically have negative ROI in Year 1 due to implementation costs. Break-even typically occurs in Year 2–3. This is important context for management approval decisions.

6. Resistance to MIS Adoption – Kotter's 8-Step Change Model

One of the biggest MIS implementation failures is not technical — it's human. Employees resist change for many reasons: fear of job loss, unfamiliarity with technology, disruption of established routines, or lack of perceived benefit. John Kotter's 8-Step Change Model provides a structured approach:

○ Step 1 – Create a Sense of Urgency

Show the organization why the current state is unsustainable. Example: "Our competitors using ERP fulfill orders in 2 days; we take 5 days. We will lose market share without MIS."

○ **Step 2 — Build a Guiding Coalition**

Assemble a team of influential leaders across departments who champion the MIS initiative — not just IT staff, but respected managers from Finance, Operations, and HR.

○ **Step 3 — Form a Strategic Vision**

Create a clear, compelling vision: "By deploying MIS, we will cut order processing time by 60%, reduce inventory costs by 20%, and enable real-time performance monitoring for all managers."

○ **Step 4 — Enlist a Volunteer Army**

Communicate the vision widely. Early adopters and enthusiastic users become internal advocates who help convince skeptics.

○ **Step 5 — Enable Action by Removing Barriers**

Address obstacles: provide training, redesign workflows that conflict with MIS, address IT infrastructure gaps, and reassign resistant managers.

○ **Step 6 — Generate Short-Term Wins**

Identify and publicize early successes. "The new inventory module saved Department A 13 lakh in the first month." This builds credibility and momentum.

○ **Step 7 — Sustain Acceleration**

Don't declare victory too early. Expand MIS to more departments, add features, and continuously improve based on user feedback.

○ **Step 8 — Institute Change**

Make MIS usage the new normal. Update job descriptions, performance reviews, and SOPs to incorporate MIS. The change is permanent when it becomes "how we work."

7. Case Study — ERP Implementation in a Manufacturing Firm

CASE STUDY

Berger Paints Bangladesh: ERP-Driven MIS Transformation

Background: Berger Paints Bangladesh Limited, one of the country's largest paint manufacturers, operates factories, depots, and distribution networks across Bangladesh. Prior to ERP implementation, each function (Finance, Sales, Inventory, HR) maintained separate systems with no integration.

Problem: Month-end financial closing took 15 days because accounts had to be manually reconciled across systems. Inventory data was always 1–2 days old. Sales managers couldn't see real-time stock availability when taking orders, leading to over-promising and customer dissatisfaction.

MIS Solution: Implementation of SAP ERP — an integrated MIS platform. All departments now operate on a single system. A sale entered by the sales team immediately updates inventory, triggers a dispatch order, updates accounts receivable, and is visible on the management dashboard.

Results:

- Month-end closing reduced from 15 days to 2 days
- Inventory accuracy improved from ~85% to ~99%
- Order fulfillment lead time reduced by 40%
- Management now reviews real-time dashboards instead of waiting for weekly paper reports

Lesson: MIS doesn't just automate existing processes — it transforms how an organization operates and makes decisions. The biggest benefit was not cost savings but decision speed and accuracy.

System Concepts & Information Systems Defined

CO1

Bloom's: Understand

Teaching: Lecture + Visual + Q&A

1. System Definition

DEFINITION

A **system** is a set of interrelated and interacting components that work together toward a common goal, producing outputs from inputs through defined processes, and adjusting behaviour based on feedback.

The concept of a system is foundational to understanding MIS. Before understanding an *information* system, one must understand what a system is in general.

2. System Components

Input

The resources, data, or energy entering the system from the environment.

Example: raw materials into a factory; customer orders into a sales system; sensor readings into a monitoring system.

Process

The transformation mechanism that converts inputs into outputs.

Example: manufacturing operations; data processing algorithms; financial calculations in an accounting system.

Output

The products, services, or information produced by the system for its environment.

Example: finished goods; management reports; alerts and notifications; customer invoices.

Feedback

Information about system output fed back to compare against goals, enabling adjustment.

Example: a thermostat reads room temperature (output) and adjusts heating (process) to maintain the target (goal).

Control

The mechanism that uses feedback to regulate the system, keeping it aligned with its goal.

Example: quality control inspectors stopping a production line when defect rate exceeds threshold.

SYSTEM COMPONENTS — WORKED EXAMPLE (UNIVERSITY REGISTRATION SYSTEM)

Input: Student applications, course capacities, faculty schedules, fee payment confirmations.

Process: Validate eligibility → allocate seats → generate timetables → issue registration confirmation.

Output: Registered course list per student, timetable, fee receipt, class rosters for faculty.

Feedback: After Week 1, enrollment data shows Course X is 150% over-subscribed. System flags this.

Control: Registrar opens an additional section of Course X (adjusts process/input based on feedback).

3. Open vs. Closed Systems

Dimension	Open System	Closed System
Environment interaction	Exchanges inputs/outputs with the environment	No interaction with the environment — self-contained
Adaptability	Adapts to environmental changes	Cannot adapt — fixed behaviour
Real-world examples	Almost all organizational systems (businesses, universities, hospitals)	Rare in practice; some embedded control systems approximate this
MIS relevance	All MIS are open systems — they receive external data (market prices, customer orders) and produce outputs that affect the environment	A purely mechanical timer circuit — no external data input affects its operation

4. Sub-systems and Supra-systems

- ▶ **Sub-system:** A component system within a larger system. Example: the Payroll sub-system is part of the larger HR-MIS, which is itself part of the enterprise MIS.
- ▶ **Supra-system (Super-system):** The larger system of which the current system is a part. Example: the national banking system is the supra-system for an individual bank's MIS.
- ▶ **System boundary:** The conceptual line dividing what is inside the system (controllable) from what is in the environment (external, often uncontrollable).

KEY INSIGHT — WHY THIS MATTERS FOR MIS DESIGN

When designing an MIS, you must clearly define the system boundary. What data comes from outside (market prices, customer orders, government regulations) vs. what is generated internally (production data, employee records)? The boundary determines what the MIS can control and what it must simply monitor and report.

5. Information System (IS) – Definition

DEFINITION

An **Information System (IS)** is a set of interrelated components that collect (or retrieve), process, store, and distribute information to support decision making, coordination, control, analysis, and visualization in an organization.

IS is broader than software. It includes:

- ▶ **Hardware:** Computers, servers, mobile devices, sensors, printers, storage systems
- ▶ **Software:** Operating systems, application programs, ERP systems, databases
- ▶ **Data:** Databases, data warehouses, files, documents
- ▶ **People:** IT staff, managers, end users, data analysts
- ▶ **Processes:** Business procedures, policies, workflows that the IS supports
- ▶ **Networks:** Telecommunications links connecting all components

6. Socio-Technical View of IS

The socio-technical perspective recognizes that IS exists at the intersection of technology and social/organizational systems. Optimal IS performance requires both dimensions to be aligned:

Technical Subsystem

Hardware, software, networks, databases — the technological infrastructure.

Can be optimized through engineering, architecture design, performance tuning.

Social Subsystem

People, culture, skills, values, power structures, organizational policies.

Requires change management, training, incentive alignment, leadership commitment.

CRITICAL INSIGHT

Many IS failures are not technical — they are social. A technically perfect ERP system implemented without adequate user training, change management, or organizational restructuring will fail. Studies

show that 60–70% of large IS projects fail to meet their objectives — and the primary reason is failure to address the social subsystem, not the technical one.

7. IS Dimensions

- ▶ **Organizational dimension:** IS must align with organizational hierarchy, culture, business processes, and politics. A centralized organization needs different IS than a decentralized one.
- ▶ **Managerial dimension:** Managers must understand IS enough to direct its development and use its outputs. IS is a tool for managerial decision-making — not just a technical concern for the IT department.
- ▶ **Technological dimension:** The hardware, software, networking, storage, and data management technologies that form the IS infrastructure.

8. How IS Supports Business Functions

Business Function	IS Support Role	Example System
Operations	Automate transactions, monitor production, manage inventory	ERP, Manufacturing Execution System (MES), TPS
Management	Provide performance reports, exception alerts, budget tracking	MIS dashboards, Management Reporting Systems
Strategy	Support long-term planning with trend analysis, competitive intelligence, scenario modelling	DSS, Business Intelligence (BI) platforms, EIS
Knowledge work	Support document creation, collaboration, professional analysis	Office Automation Systems, Knowledge Management Systems

1. Information Technology (IT) – Definition

DEFINITION

Information Technology (IT) refers to the hardware, software, and networking infrastructure used to create, process, store, secure, and exchange electronic data. IT is the *technical foundation* — the tools and infrastructure — not the system built on top of them.

IT components include:

- ▶ **Hardware:** Servers, desktop computers, laptops, tablets, smartphones, routers, switches, storage arrays, IoT sensors
- ▶ **Software:** Operating systems (Windows Server, Linux), databases (MySQL, Oracle), programming languages, middleware
- ▶ **Networking:** LAN, WAN, Internet, Wi-Fi, fiber optic links, VPNs, firewalls
- ▶ **Data storage:** Hard drives, SSDs, cloud storage, backup systems

2. IS vs. IT – Comparison

Dimension	Information Technology (IT)	Information System (IS)
Definition	Hardware + Software + Networks — the technical infrastructure	IT + People + Processes + Data — the complete system for organizational problem-solving
Scope	Technical — what machines and software can do	Socio-technical — what people and technology do together to solve business problems
Focus	Technology capabilities and infrastructure management	Business value creation and decision support
Lifecycle	Technology product lifecycle (hardware refresh every 3–5 years)	Business system lifecycle (aligns with organizational strategy and change)
Stakeholders	IT engineers, system administrators, network engineers	Business managers, end users, IT specialists, executives
Success measure	Uptime %, processing speed, storage capacity, security compliance	Decision quality, business process improvement, ROI, user adoption

Dimension	Information Technology (IT)	Information System (IS)
Example	Installing a high-performance database server with 99.99% uptime	Using that server in a CRM system that reduces customer churn by 15%
Relationship	IT is a subset/component of IS	IS is broader — encompasses IT plus the human and organizational elements

EXAM TIP

The most common exam mistake is treating IS and IT as synonyms. Remember: IT = tools (infrastructure); IS = tools + people + processes (system). A hospital buying 50 new computers is an IT investment. A hospital deploying an Electronic Medical Records system that changes how doctors work and improves diagnosis accuracy is an IS project.

3. Expanding Role of IS — Historical Evolution

○ 1950s–60s — Electronic Data Processing (Back-Office Automation)

IS began as data processing machines for accounting tasks — payroll, invoicing, record-keeping. Mainframe computers processed batch jobs overnight. IS was purely an operational efficiency tool, managed by specialists, invisible to most employees. No strategic role.

○ 1970s–80s — Management Information Systems & DSS (Management Support)

IS moved beyond back-office to support management decisions. MIS generated scheduled reports for middle managers. Decision Support Systems (DSS) emerged, allowing managers to run "what-if" analyses. IS began to influence tactical decision-making.

○ 1990s — ERP & Front-Office CRM (Enterprise-Wide Integration)

Enterprise Resource Planning (ERP) systems (SAP, Oracle) integrated all business functions into a single IS. Customer Relationship Management (CRM) systems extended IS to customer-facing roles. IS became the operational backbone of large organizations.

○ 2000s — Internet Era (Networked IS & E-Commerce)

The Internet enabled IS to connect across organizations — supply chain IS, e-commerce platforms, inter-organizational systems. IS became a competitive weapon. Amazon's IS became its primary competitive advantage over traditional retailers.

○ 2010s–Present — Cloud, Mobile, AI, IoT (Strategic Decision Tools)

IS now embedded in every aspect of business and personal life. Cloud computing enables IS without infrastructure investment. Mobile IS extends decision support to anywhere/anytime. AI-powered IS can now make or recommend decisions autonomously. IoT generates real-time data from physical assets.

4. IS and Globalization

IS has been the primary enabler of economic globalization:

- ▶ **24/7 distributed operations:** A Bangladesh software firm can receive requirements from a US client at midnight, develop code overnight, and deliver it by morning — enabled by IS and the Internet.
- ▶ **Global supply chains:** Apple's supply chain spans 43 countries. This is only manageable because IS enables real-time visibility into supplier inventory, production status, and shipping across the globe.
- ▶ **Currency and time zone management:** Global IS must handle multiple currencies, languages, time zones, and regulatory environments — an enormous technical and organizational challenge.
- ▶ **Virtual teams:** IS tools (video conferencing, collaborative platforms, shared databases) enable teams spread across continents to work as if co-located.

5. Digital Transformation — Cloud, Mobile, AI, IoT

Cloud Computing

IS delivered as a service over the Internet, eliminating the need for on-premise infrastructure. Small organizations can now afford enterprise-grade IS.

Examples: Google Workspace (Office IS), Salesforce (CRM), AWS hosting MIS applications

Mobile IS

IS accessible on smartphones and tablets — extending decision support to field workers, sales teams, and executives on the move.

Examples: Mobile ERP apps, field service management, real-time delivery tracking

Artificial Intelligence

AI enables IS to learn from data, recognize patterns, and make recommendations or autonomous decisions.

Examples: fraud detection IS, demand forecasting, chatbot-based customer service IS, recommendation engines

Internet of Things (IoT)

Physical devices (sensors, machines, vehicles) generate real-time data fed into IS — enabling unprecedented operational visibility.

Examples: smart factory sensors feeding production IS, GPS-tracked delivery vehicles, smart inventory shelf sensors

6. Emerging IS Roles — CIO and CDO

Role	Full Title	Primary Responsibility	Reports To
CIO	Chief Information Officer	Aligning IS strategy with business strategy; overseeing IS development and operations; managing IT budget and staff	CEO / Board

Role	Full Title	Primary Responsibility	Reports To
CDO	Chief Digital Officer	Leading digital transformation initiatives; driving adoption of cloud, AI, mobile, and data analytics; reimagining business models through digital technology	CEO / Board
CISO	Chief Information Security Officer	Protecting IS from cyber threats; managing information security policy; ensuring compliance with security standards (ISO 27001)	CIO / CEO
CDO (Data)	Chief Data Officer	Governing organizational data as a strategic asset; ensuring data quality, accessibility, and ethical use	CEO / CIO

1. IS Activities — The Five Core Activities

Every information system performs five fundamental activities, regardless of its type or scale:

1. Input

Capturing raw data from internal or external sources and entering it into the IS for processing.

Examples: keyboard entry, barcode scanning, sensor readings, API data feeds, file uploads, form submissions.

2. Processing

Transforming, calculating, classifying, sorting, and manipulating input data to produce meaningful information.

Examples: calculating employee net pay, generating inventory totals, running a demand forecast model, sorting records by date.

3. Output

Distributing processed information to users in a useful format — reports, screen displays, documents, or automated actions.

Examples: printed invoices, on-screen dashboards, emailed reports, automated purchase orders triggered by low stock.

4. Storage

Retaining data and information in organized form for future retrieval, analysis, and historical comparison.

Examples: relational databases, data warehouses, document management systems, cloud storage, backup archives.

5. Control

Monitoring IS performance to ensure it is operating correctly, securely, and in alignment with organizational goals.

Examples: access control (who can read/write), audit logs, error handling, backup verification, performance monitoring dashboards.

2. Input Activities — Detail

- ▶ **Data capture:** The first step — collecting raw data from its source. Can be manual (keyboard), automated (sensor), or semi-automated (OCR scanning of paper forms). The accuracy and completeness of capture directly affects output quality (GIGO).

- ▶ **Coding:** Assigning standardized codes to data for efficient processing and retrieval. Example: a product is assigned SKU code "EL-4421-BLK" encoding category (Electronics), product number, and colour. Coding enables rapid lookup and avoids ambiguity (two products named "Black Phone Case" could refer to different items; their SKUs are unambiguous).
- ▶ **Validation:** Checking input data against defined rules before it enters the system (range checks, type checks, presence checks, consistency checks — covered in Class 17).
- ▶ **Encryption in transit:** When data is entered over a network (e.g., a web form, mobile app), it must be encrypted to prevent interception. TLS/SSL ensures that data captured from a user's browser is encrypted before transmission to the IS server.

3. Processing Activities — Detail

- ▶ **Arithmetic operations:** Addition, subtraction, multiplication, division applied to numeric data. Example: $\text{Total_Revenue} = \text{Quantity} \times \text{Unit_Price} - \text{Discount_Amount}$.
- ▶ **Logical operations:** Comparison and Boolean operations. Example: IF (Stock_Level < Reorder_Point) THEN trigger_purchase_order. Logical operations enable conditional processing — the IS responds differently to different data conditions.
- ▶ **Comparison operations:** Comparing values to identify anomalies, rank items, or match records. Example: comparing actual sales to budget to compute variance; matching customer name to existing database record to detect duplicates.
- ▶ **Batch vs. online processing:** See Class 17. Batch = collected and processed together at intervals; Online = processed immediately upon input.

4. Output Activities — Types of Outputs

Output Type	Description	Example
Scheduled Reports	Generated automatically at fixed intervals	Daily sales summary, weekly inventory report, monthly payroll register
Exception Reports	Generated when a condition exceeds a threshold	Alert when machine downtime > 5%; customer overdue by 60+ days
Demand Reports	Generated on-request by a user	Manager requests "Show all customers in Chattogram with >₹1 lakh credit outstanding"
Screens / Dashboards	Real-time visual display of KPIs	Operations center wall display showing live production metrics

Output Type	Description	Example
Documents	Formal output documents for internal or external use	Invoice, purchase order, payslip, shipping manifest
Automated Actions	IS directly triggers an action without human intervention	IS automatically sends reorder email to supplier when stock falls below reorder point

5. IS Resources

IS requires five categories of resources to function:

Hardware Resources

Physical devices used by the IS: computers, servers, storage devices, printers, scanners, network equipment, IoT sensors, mobile devices.

Managing hardware resources = capacity planning, hardware refresh cycles, disaster recovery infrastructure.

Software Resources

Programs and procedures that control hardware and process data: operating systems, application software, database management systems (DBMS), middleware, utilities.

Examples: Windows Server (OS), SAP (ERP application), Oracle (DBMS), Apache (web server middleware).

Data Resources

The raw material of IS — organized collections of data: databases, knowledge bases, data warehouses, document repositories, metadata repositories.

A customer database contains: name, address, contact, order history, credit limit — each a data resource.

People Resources

The human component — IS specialists who build and maintain the system, and end users who use its outputs.

IS Specialists: system analysts, programmers, database admins, network engineers, IT managers. End Users: accountants using payroll IS, sales managers using CRM, executives using dashboards.

Network Resources

Communication infrastructure connecting IS components: LAN, WAN, Internet, intranet, extranet, wireless networks, protocols.

Without network resources, IS cannot share information across departments, branches, or with external partners and customers.

6. Data Resources — Deeper Look

- ▶ **Database:** An organized collection of structured data stored in a DBMS. Each piece of data is stored once but can be accessed by multiple applications simultaneously. Example: a single customer database accessed by the CRM, billing system, and shipping system.
- ▶ **Knowledge base:** Stores structured and unstructured knowledge — rules, best practices, case histories, expert insights — for retrieval and reuse. Used in Expert Systems and Knowledge Management Systems.
- ▶ **Metadata repository:** Stores data *about* data — describing the structure, meaning, and context of other data. Example: a metadata catalog might record that "Column: CUST_DOB, Type: DATE, Format: YYYY-MM-DD, Source: Registration Form, Owner: HR Dept." This is essential for data governance and ETL processes.

7. People Resources — IS Specialists vs. End Users

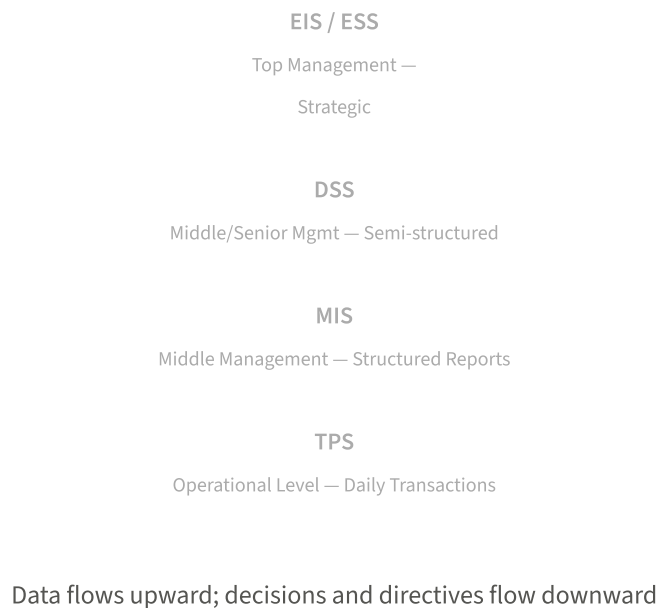
Role	Category	Responsibilities
System Analyst	IS Specialist	Bridges business needs and technical solutions; gathers requirements; designs IS architecture
Programmer / Developer	IS Specialist	Writes, tests, and maintains software components of the IS
Database Administrator (DBA)	IS Specialist	Designs, maintains, and tunes databases; manages access control and backup
Network Engineer	IS Specialist	Designs and maintains the network infrastructure supporting the IS
IT Manager / CIO	IS Specialist (Management)	Aligns IS strategy with business goals; manages IS staff and budget
Accountant using payroll IS	End User	Uses IS to perform job functions; provides feedback on IS usability and quality
Sales Manager using CRM	End User	Relies on IS outputs for decision-making; must trust and understand IS outputs
Executive using dashboard	End User (Executive)	Uses IS strategic outputs for high-level decisions; needs IS to be reliable and timely

KEY INSIGHT

IS specialists build and maintain the system; end users create value from it. The gap between these two groups — called the IS-business gap — is a major source of IS project failures. IS specialists may not understand business needs; end users may not communicate their needs clearly. Bridging this gap is the primary role of the system analyst.

1. The IS Hierarchy Pyramid

Information Systems can be classified by the management level they serve. The traditional pyramid model shows data flowing upward and decisions flowing downward:



2. Transaction Processing System (TPS)

DEFINITION

A **Transaction Processing System (TPS)** is a computerized system that processes the routine, high-volume, day-to-day transactions of an organization in a reliable, accurate, and fast manner. TPS generates the raw operational data that all higher-level IS depend on.

- ▶ **Decision type:** Structured — fully predefined rules govern every transaction. No managerial judgment required.
- ▶ **Data volume:** Very high — thousands to millions of transactions per day in large organizations.
- ▶ **Speed requirement:** High — transactions must be processed in seconds or milliseconds.
- ▶ **Users:** Operational-level staff — clerks, cashiers, warehouse staff, tellers.
- ▶ **Key feature — ACID:** Atomicity, Consistency, Isolation, Durability guarantee transaction integrity.

TPS EXAMPLES

Banking TPS: ATM withdrawal — debits account, dispenses cash, updates balance, prints receipt — all in one atomic transaction. If any step fails, the entire transaction is rolled back.

Retail TPS: Point-of-Sale (POS) system records each item scanned, calculates total, processes payment, updates inventory, prints receipt, and records the transaction in the sales database.

Airline TPS: Seat reservation system — marks a seat as booked, records passenger details, issues booking reference, updates available seat count — in real-time for all agents simultaneously.

3. Management Information System (MIS)

DEFINITION (IN THIS CLASSIFICATION CONTEXT)

At the **MIS level** in the IS hierarchy, the system takes TPS data, processes it into summarized reports, and presents it to middle managers for monitoring performance and making structured decisions.

- ▶ **Decision type:** Structured — decisions follow defined procedures (if variance > 10%, investigate; if inventory < threshold, reorder).
- ▶ **Output:** Scheduled reports, exception reports, demand reports — not raw transactions.
- ▶ **Users:** Middle management — department heads, regional managers, functional managers.
- ▶ **Time horizon:** Past and present — historical performance vs. current targets.
- ▶ **Examples:** Monthly sales performance report by region; weekly inventory status vs. reorder point; quarterly budget variance report.

4. Decision Support System (DSS)

DEFINITION

A **Decision Support System (DSS)** is an interactive IS that helps managers with semi-structured and unstructured decisions by providing analytical tools, models, and flexible access to data. DSS supports — but does not make — decisions; the manager retains control.

- ▶ **Decision type:** Semi-structured — part of the problem can be structured (data analysis), but part requires human judgment (what does this mean for our strategy?).
- ▶ **Key feature — What-If Analysis:** Managers can change assumptions and instantly see how outcomes change. Example: "What if we increase the advertising budget by 20%? What is the projected impact on

sales?"

- ▶ **Key feature — Sensitivity Analysis:** Which input variable has the most impact on the output? Example: Is our profit more sensitive to raw material price changes or to demand volume changes?
- ▶ **Key feature — Goal Seeking:** Working backwards from a desired outcome to find the required input. Example: "To achieve ₹5 crore profit, what sales volume do we need?"
- ▶ **Users:** Senior and middle managers, analysts, planners.

DSS WORKED EXAMPLE — PRICING DECISION

A supermarket chain's pricing manager faces a semi-structured decision: setting the price of a new product. The DSS provides:

- Historical price-elasticity data for similar products (structured component)
- Competitor pricing from a web scraping feed (structured component)
- A what-if model: "If we price at ₹120, projected weekly sales = 850 units; at ₹100, projected sales = 1,200 units."
- Break-even calculation: cost = ₹75/unit; at ₹120, margin = ₹45/unit × 850 = ₹38,250/week

The manager uses this DSS output — but still applies judgment: "Our strategy is volume-based market share, so I choose ₹100 despite the lower margin." DSS informs, the manager decides.

5. Executive Information System / Executive Support System (EIS/ESS)

DEFINITION

An **Executive Information System (EIS)**, also called Executive Support System (ESS), is a highly aggregated, graphical, and flexible IS designed for top executives to monitor the overall performance of the enterprise, identify strategic trends, and support unstructured strategic decisions.

- ▶ **Decision type:** Unstructured — strategic, long-horizon, high-uncertainty decisions with no predefined solution procedure.
- ▶ **Key feature — Drill-down:** Executives see high-level summaries but can drill down into details when needed. Example: CEO sees "Total Revenue: ₹450 crore" → drills down to region → drills down to product → drills down to individual transaction.
- ▶ **Key feature — Exception highlighting:** Critical variances are automatically highlighted in red/amber/green.
- ▶ **External data integration:** EIS incorporates external data — stock market prices, competitor news, macroeconomic indicators — alongside internal performance data.

- **Users:** CEO, board members, top executives, C-suite.
- **Format:** Graphical dashboards, scorecards, trend charts — minimal tabular data.

6. Knowledge Management System (KMS)

DEFINITION

A **Knowledge Management System (KMS)** is an IS that supports the capture, storage, organization, retrieval, and sharing of an organization's knowledge — both explicit (documented) and tacit (in people's heads).

- **Explicit knowledge:** Documented in manuals, procedures, reports, databases — can be stored and retrieved directly.
- **Tacit knowledge:** In people's experience, skills, and judgment — harder to capture. KMS attempts to make tacit knowledge explicit through expert interviews, communities of practice, and case documentation.
- **Examples:** Consulting firm knowledge databases (McKinsey Knowledge Center), hospital clinical decision support systems, legal research platforms (LexisNexis), corporate wikis (Confluence).

7. Office Automation System (OAS)

DEFINITION

An **Office Automation System (OAS)** supports day-to-day knowledge worker activities — communication, document creation, scheduling, and collaboration.

- **Email systems:** Microsoft Outlook, Gmail — electronic messaging replacing paper memos
- **Document management:** MS Word, Google Docs — creating, editing, storing documents electronically
- **Scheduling and calendar:** Google Calendar, MS Outlook Calendar — coordinating meetings across teams
- **Video conferencing:** Zoom, MS Teams — replacing physical meetings
- **Collaboration platforms:** Slack, MS Teams — real-time team communication and file sharing

8. Comprehensive Comparison Table — TPS vs. MIS vs. DSS vs. EIS

Dimension	TPS	MIS	DSS	EIS/ESS
Management level	Operational	Middle	Middle/Senior	Top / Executive
Decision type	Structured (routine)	Structured (reporting)	Semi-structured	Unstructured (strategic)
Data source	Live transactions	TPS summary	TPS + MIS + external	All sources + external
Output format	Transaction records, receipts	Scheduled/exception reports	What-if models, analyses	Graphical dashboards, scorecards
Time horizon	Present (real-time)	Past and present	Present and near-future	Long-term / strategic
Frequency of use	Continuous — every transaction	Regular — daily/weekly/monthly	On-demand for specific decisions	On-demand / exception-triggered
User interaction	Minimal judgment — follow procedure	Monitor and act on reports	High — "what-if" exploration	Browse, drill-down, monitor
Bangladesh example	bKash mobile money transaction	Grameenphone monthly subscriber churn report	BGMEA export market analysis tool	Bangladesh Bank Governor's economic dashboard

EXAM TIP

The classification table (TPS vs. MIS vs. DSS vs. EIS) is one of the most frequently examined topics in this course. Know all dimensions cold. A common exam question: "Distinguish between MIS and DSS with examples." Key difference: MIS produces predefined reports for structured decisions; DSS supports interactive what-if analysis for semi-structured decisions. MIS tells you what happened; DSS helps you decide what to do about it.

1. Organizational Structures and IS Fit

The structure of an organization determines how information must flow — and therefore what kind of IS is needed. Four primary organizational structures exist, each with different IS implications:

Structure	Description	IS Implication	Example
Hierarchical (Functional)	Traditional pyramid — clear chain of command, functional departments (Finance, HR, Marketing, Operations), formal reporting lines	Centralized IS with structured reporting. Each department has its own IS module; reports flow upward through the hierarchy. Strong access controls by level.	Traditional banks, government ministries, manufacturing companies like Bashundhara Group
Flat	Few or no middle management layers — decisions pushed to the lowest level. Broad spans of control.	IS must empower front-line employees with real-time data. Dashboards replace managerial filters. Faster information flow essential.	Tech startups, retail chains like Walmart (operationally flat at store level)
Matrix	Dual reporting lines — employees report to both a functional manager and a project manager simultaneously	IS must handle complex reporting relationships. Resource allocation IS critical — who is assigned to which project? Time tracking IS must serve both functional and project dimensions.	Consulting firms, advertising agencies, construction companies managing multiple projects
Network	Loose federation of independent units connected by IS and contractual relationships. Core firm outsources many functions.	IS is the glue holding the network together. Inter-organizational IS (shared databases, APIs, EDI) essential. Trust and security are critical — partners access parts of the IS.	Nike (outsources manufacturing; IS coordinates global supplier network), Uber (IS platform connecting independent drivers)

2. Centralized vs. Decentralized IS

Dimension	Centralized IS	Decentralized IS
Data location	All data stored and processed in a central location (headquarters data center or cloud)	Data stored and processed at local units (branch offices, regional data centers)
Control	High central control — consistent standards, uniform processes across the organization	Local units have autonomy — can customize IS to local needs
Cost	Lower hardware cost (economies of scale); higher network dependency	Higher hardware cost (duplicate systems); lower network dependency
Flexibility	Lower — changes must be approved and applied centrally	Higher — local units can adapt quickly
Risk	Single point of failure — if the central system goes down, the entire organization is affected	Failure is isolated — one branch's system failure doesn't affect others
Data consistency	High — single database, no synchronization needed	Lower — data may be inconsistent between units; requires synchronization
Best for	Organizations needing uniform reporting, strict compliance, cost efficiency	Organizations with diverse regional needs, high autonomy requirements, or poor network connectivity in remote locations

3. IS and Business Process Redesign (BPR)

DEFINITION

Business Process Redesign (BPR), also called Business Process Reengineering, is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical performance measures such as cost, quality, service, and speed — enabled by IS.

IS enables four levels of business process change, from least to most radical:

○ Level 1 — Automation

Using IS to perform existing manual tasks faster and with fewer errors. The process itself does not change — IS just replaces human effort. Example: Replacing manual calculation of monthly payroll with payroll software. The payroll process is identical; IS just does the arithmetic. *Low risk, limited benefit.*

○ Level 2 — Rationalization

Streamlining existing procedures before automating them — removing unnecessary steps, reducing paperwork, eliminating redundancies. Example: Before automating the loan approval process, the bank

removes 3 redundant approval layers and consolidates two review steps into one. Then automates. *Moderate improvement.*

○ Level 3 — Reengineering (BPR)

Radically redesigning the entire process from scratch, using IS as an enabler. No assumption about how the process currently works is carried forward. Example: Ford Motor Company's accounts payable reengineering: previously 500 clerks matched purchase orders, receiving documents, and invoices (a 3-way match). BPR eliminated the invoice entirely — IS automatically approves payment when goods are received and match the purchase order. Staff reduced to 125. *Dramatic improvement, high risk.*

○ Level 4 — Paradigm Shift

Reconceiving the entire nature of the business and its relationship with customers, suppliers, and competitors. IS enables entirely new business models. Example: Traditional book retailers → Amazon (online IS-driven retail with no physical stores, infinite inventory, algorithmic recommendations, 1-click purchasing). The entire industry paradigm shifted. *Transformational, highest risk and reward.*

4. How IS Changes the Nature of Work

- ▶ **Deskilling:** IS can reduce the skill required for certain tasks by automating complex procedures. A cashier no longer needs to memorize prices — the POS system looks them up automatically. An accountant no longer needs to know how to calculate depreciation — the accounting IS applies the formula. Risk: workforce becomes dependent on IS; if it fails, staff cannot perform basic functions.
- ▶ **Upskilling:** Paradoxically, IS also creates demand for higher-order skills. Workers freed from routine tasks must now analyze IS outputs, interpret data, and make more complex decisions. Example: accountants now spend less time on data entry and more time on financial analysis and strategic advice.
- ▶ **Job displacement:** IS automation reduces the number of people needed for certain roles. Bank tellers displaced by ATMs; travel agents displaced by online booking IS; call centre agents displaced by AI chatbots.
- ▶ **New job creation:** IS creates entirely new roles that did not exist before: data analysts, UX designers, cybersecurity specialists, cloud architects, digital transformation managers.

5. Virtual Organizations

DEFINITION

A **virtual organization** is an organization that uses IS and telecommunications to connect geographically dispersed employees, contractors, and partners, enabling them to work together as if they were in a single location — without a fixed physical presence.

- ▶ **IS requirements for virtual organizations:** Cloud-based IS, video conferencing, shared project management platforms (Jira, Asana), real-time document collaboration (Google Workspace), VPN-secured remote access, time zone management tools.
- ▶ **Advantages:** Access to global talent pool; lower real estate costs; 24/7 operations across time zones; rapid scaling without physical infrastructure.
- ▶ **Challenges:** Communication gaps due to time zones and cultural differences; maintaining organizational culture virtually; cybersecurity risks of distributed access; performance monitoring without physical observation.
- ▶ **Bangladesh example:** Many Bangladeshi software companies (Therap BD, Brain Station 23) operate with remote development teams serving global clients — a virtual organization model enabled entirely by IS.

6. IS and Organizational Culture

- ▶ **IS as a cultural mirror:** The IS an organization deploys reflects its culture. A command-and-control culture builds IS with rigid hierarchical access and top-down reporting. A collaborative culture builds IS with open data sharing and bottom-up input.
- ▶ **Resistance:** IS implementation often encounters cultural resistance. If the culture values privacy and autonomy, employees resist IS that monitors their productivity in real-time. If the culture is siloed, departments resist sharing data with a centralized IS.
- ▶ **Power shifts:** IS redistributes information — and therefore power. When a factory floor supervisor can see real-time production data that previously only the plant manager saw, power dynamics shift. Middle managers may resist IS that gives top management direct visibility into operations, reducing the value of the middleman's information filtering role.
- ▶ **Change management:** Successful IS implementation requires managing the cultural change alongside the technical change (refer to Kotter's 8-Step model from Class 18).

7. Case Study — Walmart's IS and Flat Supply Chain Management

CASE STUDY

Walmart: How IS Enables a Flat, Decentralized Supply Chain at Global Scale

Background: Walmart is the world's largest retailer, with over 10,000 stores across 24 countries and annual revenues exceeding \$600 billion. Its competitive advantage is rooted not in its products — which are the same as competitors — but in its IS-driven supply chain efficiency.

Organizational Challenge: Walmart operates a relatively flat organizational structure at the store level — store managers have significant autonomy over inventory and staffing decisions. Yet the company needs central

coordination of purchasing, pricing, and supplier relationships across thousands of locations. This is a classic centralized vs. decentralized tension.

IS Solution: Retail Link — Walmart's proprietary supplier information system. Retail Link gives suppliers real-time access to their product's sales data across all Walmart stores globally. A soap manufacturer in Chittagong supplying Walmart USA can see exactly how many units sold in each store today, current inventory levels, and projected reorder quantities — without any communication from Walmart's procurement team.

How IS enables the flat structure:

- **Decentralized execution:** Store managers use local IS dashboards to make real-time inventory decisions without waiting for headquarters approval. IS empowers local decision-making.
- **Centralized coordination:** The Retail Link IS simultaneously gives central procurement teams visibility into what every supplier is delivering and what every store is selling — enabling coordinated pricing and purchasing strategies without hierarchical bottlenecks.
- **Cross-docking IS:** Walmart's IS coordinates a just-in-time logistics system where goods from suppliers are transferred directly from inbound to outbound trucks at distribution centers — reducing warehouse storage time from days to hours. This required IS that coordinates supplier shipment schedules, distribution center operations, and store receiving in real-time.

Results: Walmart's inventory turnover is approximately 2× the industry average. Its cost of goods is 5–10% lower than competitors due to supply chain IS efficiency. This cost advantage is passed to customers as lower prices — the IS-driven competitive strategy that has transformed global retail.

MIS Lesson: IS can enable an organization to simultaneously achieve the benefits of both centralization (coordinated purchasing, uniform data, economies of scale) and decentralization (local flexibility, speed, empowered front-line decision-making) — something impossible with purely hierarchical, paper-based management systems.